Supporting Information

Aridity and herbivory shape demographic benefits of fungal endophytes in cool-season grasses

Jacob K. Moutouama, Joshua C. Fowler, Akiem M. Gough,
Julia Martin, Christopher C. Oxley, Karl M. Schrader, Karis Williams, and Tom E.X. Miller

S.1 Supporting Information

S.1.1 Supporting Methods

S.1.1.1 Endophyte status verification

To assess the efficacy of heat treatment and to confirm endophyte status before transfer to the field experiment, we verified symbiont status (S^+ or S^-) of plants from control and heat lineages using three methods. First, peels of the inner leaf sheaths were stained with aniline blue lactic acid and viewed under a compound microscope at 200×–400× to identify *Epichloë* hyphae (Jackson and Johnson-Cicalese, 1988). Second, we used microscopy to assess the presence or absence of *Epichloë* in seeds that were soaked in NaOH, then squashed and stained similarly to the leaf tissue, to determine the endophyte status of the maternal plant (Fowler et al., 2025). Third, we used an immunoblot assay (Phytoscreen field tiller endophyte detection kit, Agrinostics Ltd. Co., Watkinsville, Georgia, USA) that targets proteins of *Epichloë* spp. to visually indicate presence or absence (Koh et al., 2006). In total, we tested 1,481 plants for endophyte status. In cases where two or more methods were used on the same plants ($N = 253$), there was 99.2% agreement between methods. For genotypes that were vegetatively propagated, we tested the endophyte status of a single clonal replicate and assumed that its status applied equally to all clones.

All S^- plants used in the experiment came from heat-treated lineages, while S^+ plants came from both control and heat lineages depending on the efficacy of the removal treatment. While heat treatment was effective at reducing endophyte prevalence relative to controls, our confidence that individual S^- plants were experimentally disinfected, as opposed to naturally symbiont-free, varied due to among-population variation in endophyte prevalence (mean: 68% S^+) and the effectiveness of the heat treatment (Table S2). For example, for the SJC population of *A. hyemalis*, control and heat lineages yielded plants that were 100% and 5% S^+, respectively – giving us high confidence that S^- plants are derived from successful symbiont removal. For the SFAEF population of *P. autumnalis*, on the other hand, control and

heat lineages yielded plants that were 35% and 26% S^+, respectively – meaning that these S^- plants were quite likely from naturally S^- lineages.

S.1.2 Greenhouse experiment

S.1.2.1 Experimental design

To examine whether edaphic factors mediate plant performance across the precipitation gradient, we conducted a greenhouse experiment manipulating soil origin and endophyte status. Soils were collected during summer 2024 from the seven field sites used in the common garden experiment. These sites span the precipitation gradient from the wettest site to the driest site. After collection, soils were stored in a cold room at 3.3°C until the start of the experiment. Soils were not sterilized and therefore soil origin treatment reflects both the physical properties of the soil as well as the microbial communities that may vary across soil types.

Seeds of *Agrostis hyemalis* were obtained from the seed stock used to generate plants in the range-limits field experiment. Seeds originated from two source populations (LPEF and SJC) and were grouped according to endophyte status as endophyte-symbiotic (S^+) or endophyte-free (S^-). Seeds were germinated on agar plates (18.2 g agar per 1000 mL deionized water) in a growth chamber and subsequently transplanted as seedlings into pots containing soils from each field site. Plants were randomly assigned to greenhouse trays and maintained under controlled greenhouse conditions with automated irrigation. The experimental design crossed seed source population, endophyte status, and soil origin. Realised sample sizes varied across treatment combinations due to mortality and poor germination and are reported in Table S5.

S.1.2.2 Data collection

Plants were monitored regularly for survival and reproductive development. Reproductive output was quantified as inflorescence number. During the flowering season, plants were checked at least twice per week for the presence of seed-bearing inflorescences. Mature inflorescences were collected and counted as they developed to avoid repeated counts. After eight months of growth, plants were harvested to quantify aboveground biomass. Plant material was dried at 60°C in a drying oven and weighed using an analytical balance. Because inflorescences were collected prior to harvest, they were dried separately and added to the total aboveground biomass.

S.1.2.3 Statistical analysis

For each fitness component (aboveground biomass and inflorescence count), we fitted hierarchical Bayesian models mirroring the field analysis. Fixed effects included symbiont

status (S^+/S^-), the 30-year mean annual precipitation of each soil's origin site (standardized; hereafter soil-origin MAP), and their interaction, with planting date (standardized) as a covariate. Source population was included as a random effect with a non-centered parameterization. Biomass was log-transformed and modeled with a Student-t likelihood ($\nu = 3$) to provide robustness to outliers, which are common in small greenhouse experiments; inflorescence count was modeled with a negative binomial likelihood to accommodate overdispersion. For all models, fixed-effect coefficients received Normal$(0,1)$ priors; the overdispersion parameter for the inflorescence model was placed on the log scale with a Normal$(2,1)$ prior (centered on $\phi \approx 7$); and the random-effect standard deviation received an Exponential(2) prior.

Models were fitted in Stan (Stan Development Team, 2024) with four chains of 2,000 iterations (1,000 warmup; adapt_delta $= 0.99$). Convergence was assessed via $\hat{R} < 1.01$ for all parameters and inspection of trace plots. We report posterior medians with 95% credible intervals for three focal parameters: the symbiont main effect (S^+ vs. S^- at mean soil-origin MAP), the soil-origin MAP main effect, and the symbiont $\times$ soil-origin MAP interaction. To visualize how the endophyte effect varies across the soil-origin precipitation gradient, we computed the posterior contrast ($S^+ - S^-$) at each point along the observed precipitation range, marginalizing over planting date (set to the mean) and source population random effects. These contrasts are summarized as posterior medians with 95% credible intervals and displayed alongside the predicted responses for each symbiont status. Posterior predictive checks indicated adequate fit for both models (Fig. S3).

Supporting Tables

Table S1: Geographic coordinates and descriptions of source populations for study species. For each source population, the table shows the total number of plants included in the experiment, the number of unique genotypes, and the average clonal replication per genotype.

Species	Population	Latitude	Longitude	Notes	Total plants	Unique genotypes	Mean clones per genotype
A. hyemalis	SJC	30.7720	−95.2006	Private property, San Jacinto County, TX	273	201	1.36
A. hyemalis	COHR	29.6728	−96.5675	Roadside collection, Columbus, TX	289	35	8.26
A. hyemalis	LPEF	30.0854	−97.1725	Stengl Lost Pines Field Station, Bastrop, TX	278	41	6.78
P. autumnalis	LA7	30.8497	−93.2768	Roadside population, DeRidder, LA	91	9	10.11
P. autumnalis	LA1	32.5681	−93.9330	Walter P. Jacobs Nature Center, Shreveport, LA	129	7	18.43
P. autumnalis	LA2	31.1830	−92.6170	Kisatchie National Forest, LA	253	15	16.87
P. autumnalis	SFAEF	31.4922	−94.7722	Stephen F. Austin Experimental Forest, TX	247	41	6.02
E. virginicus	HUNT	30.7418	−95.4745	Pineywoods Environmental Research Laboratory, Huntsville, TX	91	61	1.49
E. virginicus	RL5	30.4717	−99.7838	Llano River Field Station, Junction, TX	131	91	1.44
E. virginicus	PALM	29.5916	−97.5848	Palmetto State Park, TX	442	195	2.27
E. virginicus	JLP	29.7851	−90.1149	Jean Lafitte Park, LA	176	102	1.73

Table S2: Summary of symbiont removal treatments by species and population. N indicates the number of plants tested for symbiont status and $\%S^+$ indicates the proportion that tested positive. Treatment refers to the symbiont removal treatment applied to the seed lineage: Control (no treatment) or Heat (5 days in a drying oven at 60°C for *E. virginicus* and *P. autumnalis*; 12 min in a 60°C water bath for *A. hyemalis*). For all populations except SFAEF and PALM, heat-treated seed lineages were one or more generations removed from the treatment; for those two populations, plants from heat-treated seeds were used directly in the experiment.

Species	Population	Treatment	N	%S+
A. hyemalis	COHR	Control	84	0.92
A. hyemalis	COHR	Heat	75	0.04
A. hyemalis	LPEF	Control	7	0.57
A. hyemalis	LPEF	Heat	88	0.00
A. hyemalis	SJC	Control	132	1.00
A. hyemalis	SJC	Heat	191	0.05
E. virginicus	HUNT	Control	71	0.61
E. virginicus	HUNT	Heat	38	0.21
E. virginicus	JLP	Control	27	1.00
E. virginicus	JLP	Heat	113	0.02
E. virginicus	PALM	Control	176	0.61
E. virginicus	PALM	Heat	238	0.71
E. virginicus	RL5	Control	68	0.43
E. virginicus	RL5	Heat	74	0.73
P. autumnalis	LA1	Control	5	0.20
P. autumnalis	LA1	Heat	4	0.00
P. autumnalis	LA2	Control	10	0.80
P. autumnalis	LA2	Heat	3	0.33
P. autumnalis	LA7	Control	5	1.00
P. autumnalis	LA7	Heat	11	0.00
P. autumnalis	SFAEF	Control	23	0.35
P. autumnalis	SFAEF	Heat	38	0.26

Table S3: Common garden sites used in the field experiment, ordered from west to east.

Site	Code	Latitude	Longitude
TAMU Sonora Station	SON	30.273467	-100.561463
Private property, Kerrville, TX	KER	30.028437	-99.142888
Brackenridge Field Laboratory	BFL	30.284866	-97.780675
UT Stengl Lost Pines Field Station	BAS	30.092343	-97.172442
TAMU Ecology and Natural Resource Teaching Area	COL	30.569302	-96.366324
Center for Biological Field Studies	HUN	30.742815	-95.476881
University of Louisiana Ecology Center	LAF	30.304773	-92.012401

Table S4: Comparison of candidate models (quadratic vs. linear) for each vital rate based on leave-one-out cross-validation (LOOCV). ΔELPD is the difference in expected log pointwise predictive density relative to the best model.

Vital rate	Model type	ΔELPD	SE
Survival	Linear	0.000	0.000
Survival	Quadratic	-0.282	0.663
Growth	Linear	0.000	0.000
Growth	Quadratic	-0.078	0.430
Inflorescence	Linear	0.000	0.000
Inflorescence	Quadratic	-0.689	0.516
Spikelet	Linear	0.000	0.000
Spikelet	Quadratic	-0.332	0.579

Table S5: Sample sizes for each combination of seed source population, soil-origin site, and endophyte treatment in the greenhouse experiment. The experiment crossed two seed source populations (LPEF and SJC), seven soil-origin sites along the precipitation gradient, and two symbiont treatments (S^- and S^+). Most treatment combinations included seven replicates, except LPEF $\times$ BFL $\times$ S^- where $n = 6$ due to early mortality.

Response	Site	S-	S+
Biomass			
	LAF	14	14
	SON	14	14
	KER	14	14
	BFL	13	14
	BAS	14	14
	COL	14	14
	HUN	14	14
Inflorescence			
	LAF	14	14
	SON	14	14
	KER	14	14
	BFL	14	14
	BAS	14	14
	COL	14	14
	HUN	14	14
Spikelets			
	LAF	7	10
	SON	3	1
	KER	1	0
	BFL	3	8
	BAS	7	6
	COL	2	2
	HUN	5	12

Table S6: Sensitivity of estimated endophyte effects to exclusion of sites KER and SON, which were destroyed by deer and subsequently replanted in 2024. For each combination of vital rate (Trait), species, herbivory treatment (Access = herbivory-access plots, Exclusion = herbivore-exclusion plots), and precipitation (Precip., in mm; three values spanning each species' observed precipitation range), the table reports the posterior median and 90% credible interval [Med. Δ (90% CI)] and the posterior probability of a positive endophyte effect [P(Δ > 0)] for the full dataset and for the dataset excluding KER and SON. The endophyte effect Δ is defined as Pr(S+) − Pr(S-) for survival, or the analogous difference on the response scale (S+ − S-) for growth, inflorescence production, and spikelet production. Spikelet production was not modeled for *A. hyemalis* because spikelets in this species were too numerous and small to count reliably (see Methods). Posterior estimates were nearly identical between the full dataset and the dataset excluding KER and SON across all species, vital rates, and herbivory treatments, indicating that the disturbance and replanting at these two sites did not meaningfully influence our conclusions.

				Full data		Excl. KER/SON	
Trait	Species	Herbivory	Precip. (mm)	Med. Δ (90% CI)	P(Δ>0)	Med. Δ (90% CI)	P(Δ>0)
Growth	*A. hyemalis*	Access	714	0.05 [-0.37, 0.47]	0.58	0.02 [-0.40, 0.44]	0.53
Growth	*A. hyemalis*	Access	1300	0.27 [0.05, 0.51]	0.98	0.25 [0.03, 0.50]	0.97
Growth	*A. hyemalis*	Access	2472	0.49 [0.16, 0.88]	0.99	0.50 [0.16, 0.88]	0.99
Growth	*A. hyemalis*	Exclusion	714	-0.01 [-0.38, 0.37]	0.48	-0.01 [-0.39, 0.38]	0.49
Growth	*A. hyemalis*	Exclusion	1300	0.14 [-0.08, 0.35]	0.85	0.13 [-0.09, 0.35]	0.84
Growth	*A. hyemalis*	Exclusion	2472	0.29 [-0.07, 0.65]	0.90	0.29 [-0.09, 0.65]	0.90
Growth	*E. virginicus*	Access	633	0.04 [-0.41, 0.47]	0.56	-0.03 [-0.49, 0.44]	0.46
Growth	*E. virginicus*	Access	1213	0.17 [-0.05, 0.40]	0.90	0.15 [-0.09, 0.38]	0.85
Growth	*E. virginicus*	Access	2433	0.32 [0.05, 0.60]	0.97	0.33 [0.06, 0.62]	0.98
Growth	*E. virginicus*	Exclusion	633	0.09 [-0.34, 0.53]	0.63	0.08 [-0.37, 0.52]	0.62
Growth	*E. virginicus*	Exclusion	1213	0.10 [-0.11, 0.30]	0.78	0.09 [-0.12, 0.30]	0.77
Growth	*E. virginicus*	Exclusion	2433	0.10 [-0.16, 0.37]	0.74	0.10 [-0.16, 0.37]	0.73
Growth	*P. autumnalis*	Access	873	0.04 [-0.33, 0.40]	0.57	0.02 [-0.36, 0.39]	0.53
Growth	*P. autumnalis*	Access	1443	0.07 [-0.16, 0.27]	0.70	0.06 [-0.16, 0.26]	0.69
Growth	*P. autumnalis*	Access	2472	0.10 [-0.20, 0.38]	0.70	0.11 [-0.19, 0.39]	0.74
Growth	*P. autumnalis*	Exclusion	873	0.17 [-0.21, 0.58]	0.78	0.18 [-0.21, 0.59]	0.77

Continued on next page

Table S6 – continued from previous page

				Full data		Excl. KER/SON	
Trait	Species	Herbivory	Precip. (mm)	Med. Δ (90% CI)	P(Δ>0)	Med. Δ (90% CI)	P(Δ>0)
Growth	*P. autumnalis*	Exclusion	1443	-0.02 [-0.22, 0.20]	0.45	-0.02 [-0.23, 0.20]	0.45
Growth	*P. autumnalis*	Exclusion	2472	-0.22 [-0.56, 0.10]	0.13	-0.23 [-0.56, 0.10]	0.13
Inflorescence	*A. hyemalis*	Access	714	-0.09 [-0.59, 0.18]	0.25	-0.09 [-0.60, 0.17]	0.23
Inflorescence	*A. hyemalis*	Access	1300	0.06 [-0.45, 0.62]	0.59	0.05 [-0.42, 0.58]	0.58
Inflorescence	*A. hyemalis*	Access	2472	1.44 [-0.21, 6.22]	0.92	1.47 [-0.14, 6.27]	0.93
Inflorescence	*A. hyemalis*	Exclusion	714	-0.07 [-0.74, 0.49]	0.38	-0.06 [-0.72, 0.49]	0.40
Inflorescence	*A. hyemalis*	Exclusion	1300	0.20 [-0.47, 1.05]	0.71	0.20 [-0.46, 1.01]	0.71
Inflorescence	*A. hyemalis*	Exclusion	2472	1.66 [-0.69, 7.73]	0.88	1.70 [-0.68, 8.07]	0.88
Inflorescence	*E. virginicus*	Access	633	-0.09 [-0.58, 0.17]	0.25	-0.08 [-0.55, 0.14]	0.22
Inflorescence	*E. virginicus*	Access	1213	-0.01 [-0.33, 0.26]	0.46	-0.03 [-0.32, 0.24]	0.43
Inflorescence	*E. virginicus*	Access	2433	0.27 [-0.25, 1.44]	0.82	0.31 [-0.25, 1.50]	0.83
Inflorescence	*E. virginicus*	Exclusion	633	0.03 [-0.81, 1.06]	0.53	0.01 [-0.74, 1.02]	0.52
Inflorescence	*E. virginicus*	Exclusion	1213	0.07 [-0.49, 0.72]	0.59	0.06 [-0.48, 0.71]	0.58
Inflorescence	*E. virginicus*	Exclusion	2433	0.15 [-0.91, 1.62]	0.63	0.15 [-0.96, 1.73]	0.63
Inflorescence	*P. autumnalis*	Access	873	0.01 [-0.07, 0.20]	0.66	0.01 [-0.07, 0.21]	0.65
Inflorescence	*P. autumnalis*	Access	1443	0.26 [0.01, 0.78]	0.96	0.25 [0.01, 0.80]	0.96
Inflorescence	*P. autumnalis*	Access	2472	2.95 [0.56, 11.20]	0.99	2.87 [0.52, 12.25]	0.99
Inflorescence	*P. autumnalis*	Exclusion	873	0.18 [0.01, 0.89]	0.97	0.19 [0.01, 0.96]	0.97
Inflorescence	*P. autumnalis*	Exclusion	1443	1.00 [0.36, 2.44]	1.00	1.00 [0.36, 2.52]	1.00
Inflorescence	*P. autumnalis*	Exclusion	2472	5.52 [1.19, 21.88]	1.00	5.48 [1.05, 21.92]	1.00
Spikelet	*E. virginicus*	Access	633	-2.09 [-8.18, 2.83]	0.24	-0.94 [-8.27, 6.13]	0.41

Continued on next page

Table S6 – continued from previous page

				Full data		Excl. KER/SON	
Trait	Species	Herbivory	Precip. (mm)	Med. Δ (90% CI)	P(Δ>0)	Med. Δ (90% CI)	P(Δ>0)
Spikelet	*E. virginicus*	Access	1213	-1.00 [-4.53, 2.31]	0.30	-0.19 [-4.01, 3.60]	0.47
Spikelet	*E. virginicus*	Access	2433	0.74 [-3.23, 4.96]	0.62	0.57 [-3.12, 4.57]	0.60
Spikelet	*E. virginicus*	Exclusion	633	-4.63 [-12.13, 1.94]	0.12	-5.10 [-13.33, 2.20]	0.11
Spikelet	*E. virginicus*	Exclusion	1213	-1.12 [-4.49, 2.23]	0.28	-1.12 [-4.60, 2.25]	0.28
Spikelet	*E. virginicus*	Exclusion	2433	3.20 [-0.17, 7.60]	0.94	2.96 [-0.25, 6.88]	0.93
Spikelet	*P. autumnalis*	Access	873	-0.76 [-5.08, 3.35]	0.35	-0.52 [-5.12, 3.90]	0.42
Spikelet	*P. autumnalis*	Access	1443	-0.18 [-2.60, 2.28]	0.45	-0.01 [-2.58, 2.58]	0.50
Spikelet	*P. autumnalis*	Access	2472	0.84 [-2.35, 4.34]	0.67	0.78 [-2.25, 4.28]	0.67
Spikelet	*P. autumnalis*	Exclusion	873	2.73 [-2.78, 9.74]	0.81	2.99 [-3.03, 10.46]	0.80
Spikelet	*P. autumnalis*	Exclusion	1443	3.13 [0.02, 6.92]	0.95	3.28 [0.05, 7.07]	0.95
Spikelet	*P. autumnalis*	Exclusion	2472	3.46 [-0.60, 8.59]	0.92	3.45 [-0.69, 8.42]	0.92
Survival	*A. hyemalis*	Access	714	-0.04 [-0.18, 0.02]	0.13	-0.05 [-0.18, 0.01]	0.07
Survival	*A. hyemalis*	Access	1300	-0.01 [-0.08, 0.05]	0.35	-0.02 [-0.09, 0.04]	0.26
Survival	*A. hyemalis*	Access	2472	0.06 [-0.06, 0.20]	0.81	0.07 [-0.05, 0.22]	0.84
Survival	*A. hyemalis*	Exclusion	714	-0.07 [-0.23, -0.00]	0.04	-0.06 [-0.22, -0.00]	0.05
Survival	*A. hyemalis*	Exclusion	1300	-0.03 [-0.11, 0.04]	0.25	-0.03 [-0.10, 0.04]	0.26
Survival	*A. hyemalis*	Exclusion	2472	0.08 [-0.05, 0.24]	0.86	0.08 [-0.05, 0.24]	0.84
Survival	*E. virginicus*	Access	633	-0.06 [-0.23, 0.03]	0.11	-0.05 [-0.21, -0.00]	0.03
Survival	*E. virginicus*	Access	1213	-0.01 [-0.10, 0.07]	0.38	-0.03 [-0.11, 0.04]	0.23
Survival	*E. virginicus*	Access	2433	0.10 [-0.03, 0.27]	0.89	0.12 [0.00, 0.30]	0.95
Survival	*E. virginicus*	Exclusion	633	-0.14 [-0.34, -0.01]	0.03	-0.09 [-0.30, -0.01]	0.02

Continued on next page

Table S6 – continued from previous page

				Full data		Excl. KER/SON	
Trait	Species	Herbivory	Precip. (mm)	Med. Δ (90% CI)	P(Δ>0)	Med. Δ (90% CI)	P(Δ>0)
Survival	*E. virginicus*	Exclusion	1213	-0.01 [-0.11, 0.09]	0.44	-0.01 [-0.11, 0.09]	0.41
Survival	*E. virginicus*	Exclusion	2433	0.12 [0.01, 0.33]	0.97	0.10 [0.01, 0.31]	0.98
Survival	*P. autumnalis*	Access	714	-0.01 [-0.10, 0.03]	0.20	-0.02 [-0.10, 0.01]	0.14
Survival	*P. autumnalis*	Access	1300	-0.03 [-0.12, 0.05]	0.26	-0.04 [-0.12, 0.04]	0.19
Survival	*P. autumnalis*	Access	2472	0.01 [-0.12, 0.15]	0.61	0.02 [-0.12, 0.17]	0.65
Survival	*P. autumnalis*	Exclusion	714	-0.03 [-0.19, 0.04]	0.19	-0.02 [-0.17, 0.05]	0.23
Survival	*P. autumnalis*	Exclusion	1300	-0.01 [-0.11, 0.10]	0.41	-0.01 [-0.10, 0.09]	0.44
Survival	*P. autumnalis*	Exclusion	2472	0.06 [-0.08, 0.24]	0.78	0.05 [-0.09, 0.24]	0.77

Table S7: Posterior summaries of Δ (S+ − S−) for survival across precipitation quantiles (mm, 5%, 25%, 50%, 75%, 100%) for each species and herbivore treatment. Positive median values indicate beneficial effects of symbionts; negative medians indicate costly effects. $P(\Delta > 0)$ quantifies the strength of evidence: values near 1 indicate strong support for a benefit, values near 0 indicate strong support for a cost, and values near 0.5 indicate little or no consistent effect. Herbivore treatment: Exclusion = herbivores excluded; Access = herbivores had access.

Species	Herbivore treatment	Precipitation	Median Δ	Lower 90%	Upper 90%	P(Δ > 0)
A. hyemalis	Access	714.00	-0.04	-0.18	0.02	0.13
A. hyemalis	Access	963.00	-0.03	-0.12	0.03	0.16
A. hyemalis	Access	1300.00	-0.01	-0.08	0.05	0.35
A. hyemalis	Access	1832.00	0.02	-0.06	0.12	0.70
A. hyemalis	Access	2472.00	0.06	-0.06	0.20	0.81
A. hyemalis	Exclusion	714.00	-0.07	-0.23	-0.00	0.04
A. hyemalis	Exclusion	963.00	-0.06	-0.17	0.01	0.07
A. hyemalis	Exclusion	1300.00	-0.03	-0.11	0.04	0.25
A. hyemalis	Exclusion	1832.00	0.03	-0.07	0.14	0.70
A. hyemalis	Exclusion	2472.00	0.08	-0.05	0.24	0.86
E. virginicus	Access	525.00	-0.06	-0.27	0.02	0.10
E. virginicus	Access	760.00	-0.06	-0.20	0.03	0.13
E. virginicus	Access	1101.00	-0.03	-0.12	0.06	0.29
E. virginicus	Access	1680.00	0.04	-0.06	0.15	0.77
E. virginicus	Access	2433.00	0.10	-0.03	0.27	0.89
E. virginicus	Exclusion	525.00	-0.13	-0.40	-0.01	0.02
E. virginicus	Exclusion	760.00	-0.12	-0.29	-0.01	0.04
E. virginicus	Exclusion	1101.00	-0.04	-0.14	0.06	0.24
E. virginicus	Exclusion	1680.00	0.09	-0.02	0.21	0.91
E. virginicus	Exclusion	2433.00	0.12	0.01	0.33	0.97
P. autumnalis	Access	714.00	-0.01	-0.10	0.03	0.20
P. autumnalis	Access	963.00	-0.03	-0.11	0.03	0.19
P. autumnalis	Access	1300.00	-0.03	-0.12	0.05	0.26
P. autumnalis	Access	1832.00	-0.00	-0.12	0.11	0.49
P. autumnalis	Access	2472.00	0.01	-0.12	0.15	0.61
P. autumnalis	Exclusion	714.00	-0.03	-0.19	0.04	0.19
P. autumnalis	Exclusion	963.00	-0.03	-0.15	0.06	0.24
P. autumnalis	Exclusion	1300.00	-0.01	-0.11	0.10	0.41
P. autumnalis	Exclusion	1832.00	0.04	-0.09	0.17	0.70
P. autumnalis	Exclusion	2472.00	0.06	-0.08	0.24	0.78

Table S8: Posterior summaries of Δ ($S^+ - S^-$) for growth across precipitation quantiles (mm, 5%, 25%, 50%, 75%, 100%) for each species and herbivore treatment. Positive median values indicate beneficial effects of symbionts; negative medians indicate costly effects. $P(\Delta > 0)$ quantifies the strength of evidence: values near 1 indicate strong support for a benefit, values near 0 indicate strong support for a cost, and values near 0.5 indicate little or no consistent effect.

Species	Herbivore treatment	Precipitation	Median Δ	Lower 90%	Upper 90%	P($\Delta > 0$)
A. hyemalis	Access	714.00	0.05	-0.37	0.47	0.58
A. hyemalis	Access	963.00	0.15	-0.14	0.47	0.81
A. hyemalis	Access	1300.00	0.27	0.05	0.51	0.98
A. hyemalis	Access	1832.00	0.39	0.15	0.66	1.00
A. hyemalis	Access	2472.00	0.49	0.16	0.88	0.99
A. hyemalis	Exclusion	714.00	-0.01	-0.38	0.37	0.48
A. hyemalis	Exclusion	963.00	0.06	-0.21	0.33	0.65
A. hyemalis	Exclusion	1300.00	0.14	-0.08	0.35	0.85
A. hyemalis	Exclusion	1832.00	0.22	-0.04	0.48	0.91
A. hyemalis	Exclusion	2472.00	0.29	-0.07	0.65	0.90
E. virginicus	Access	525.00	-0.00	-0.53	0.51	0.49
E. virginicus	Access	760.00	0.07	-0.31	0.44	0.64
E. virginicus	Access	1101.00	0.15	-0.10	0.40	0.85
E. virginicus	Access	1680.00	0.24	0.04	0.44	0.98
E. virginicus	Access	2433.00	0.32	0.05	0.60	0.97
E. virginicus	Exclusion	525.00	0.09	-0.42	0.61	0.60
E. virginicus	Exclusion	760.00	0.09	-0.27	0.45	0.65
E. virginicus	Exclusion	1101.00	0.09	-0.14	0.33	0.75
E. virginicus	Exclusion	1680.00	0.10	-0.08	0.28	0.82
E. virginicus	Exclusion	2433.00	0.10	-0.16	0.37	0.74
P. autumnalis	Access	873.00	0.04	-0.33	0.40	0.57
P. autumnalis	Access	1122.00	0.06	-0.23	0.31	0.63
P. autumnalis	Access	1443.00	0.07	-0.16	0.27	0.70
P. autumnalis	Access	1923.00	0.08	-0.14	0.29	0.73
P. autumnalis	Access	2472.00	0.10	-0.20	0.38	0.70
P. autumnalis	Exclusion	873.00	0.17	-0.21	0.58	0.78
P. autumnalis	Exclusion	1122.00	0.08	-0.19	0.37	0.68
P. autumnalis	Exclusion	1443.00	-0.02	-0.22	0.20	0.45
P. autumnalis	Exclusion	1923.00	-0.13	-0.36	0.11	0.19
P. autumnalis	Exclusion	2472.00	-0.22	-0.56	0.10	0.13

Table S9: Posterior summaries of Δ ($S^+ - S^-$) for inflorescence production across precipitation quantiles (mm, 5%, 25%, 50%, 75%, 100%) for each species and herbivore treatment. Positive median values indicate beneficial effects of symbionts; negative medians indicate costly effects. $P(\Delta > 0)$ quantifies the strength of evidence: values near 1 indicate strong support for a benefit, values near 0 indicate strong support for a cost, and values near 0.5 indicate little or no consistent effect.

Species	Herbivore treatment	Precipitation	Median Δ	Lower 90%	Upper 90%	P($\Delta > 0$)
A. hyemalis	Access	714.00	-0.08	-0.61	0.18	0.26
A. hyemalis	Access	963.00	-0.06	-0.54	0.30	0.37
A. hyemalis	Access	1300.00	0.06	-0.44	0.57	0.59
A. hyemalis	Access	1832.00	0.46	-0.31	1.85	0.85
A. hyemalis	Access	2472.00	1.30	-0.29	6.06	0.91
A. hyemalis	Exclusion	714.00	-0.06	-0.74	0.54	0.40
A. hyemalis	Exclusion	963.00	0.01	-0.63	0.69	0.51
A. hyemalis	Exclusion	1300.00	0.19	-0.50	1.07	0.70
A. hyemalis	Exclusion	1832.00	0.67	-0.49	2.62	0.84
A. hyemalis	Exclusion	2472.00	1.48	-0.95	7.71	0.85
E. virginicus	Access	525.00	-0.09	-0.67	0.16	0.24
E. virginicus	Access	760.00	-0.08	-0.48	0.18	0.29
E. virginicus	Access	1101.00	-0.04	-0.35	0.23	0.41
E. virginicus	Access	1680.00	0.07	-0.24	0.48	0.67
E. virginicus	Access	2433.00	0.25	-0.27	1.36	0.81
E. virginicus	Exclusion	525.00	0.03	-0.87	1.17	0.54
E. virginicus	Exclusion	760.00	0.04	-0.68	0.92	0.56
E. virginicus	Exclusion	1101.00	0.07	-0.52	0.74	0.59
E. virginicus	Exclusion	1680.00	0.09	-0.45	0.75	0.63
E. virginicus	Exclusion	2433.00	0.11	-0.97	1.40	0.60
P. autumnalis	Access	873.00	0.02	-0.07	0.20	0.68
P. autumnalis	Access	1122.00	0.07	-0.06	0.38	0.82
P. autumnalis	Access	1443.00	0.26	0.01	0.83	0.96
P. autumnalis	Access	1923.00	0.98	0.24	2.89	0.99
P. autumnalis	Access	2472.00	2.91	0.48	11.89	0.99
P. autumnalis	Exclusion	873.00	0.19	0.02	1.00	0.97
P. autumnalis	Exclusion	1122.00	0.44	0.10	1.42	1.00
P. autumnalis	Exclusion	1443.00	0.97	0.35	2.44	1.00
P. autumnalis	Exclusion	1923.00	2.37	0.75	6.35	1.00
P. autumnalis	Exclusion	2472.00	4.97	0.91	19.92	0.99

Table S10: Posterior summaries of Δ ($S^+ - S^-$) for spikelet production across precipitation quantiles (mm, 5%, 25%, 50%, 75%, 100%) for each species and herbivore treatment. Positive median values indicate beneficial effects of symbionts; negative medians indicate costly effects. $P(\Delta > 0)$ quantifies the strength of evidence: values near 1 indicate strong support for a benefit, values near 0 indicate strong support for a cost, and values near 0.5 indicate little or no consistent effect.

Species	Herbivore treatment	Precipitation	Median Δ	Lower 90%	Upper 90%	P($\Delta > 0$)
E. virginicus	Access	525.00	-2.33	-9.38	3.12	0.23
E. virginicus	Access	760.00	-1.84	-7.08	2.69	0.24
E. virginicus	Access	1101.00	-1.19	-4.99	2.31	0.28
E. virginicus	Access	1680.00	-0.27	-3.26	2.81	0.44
E. virginicus	Access	2433.00	0.74	-3.23	4.96	0.62
E. virginicus	Exclusion	525.00	-5.51	-14.68	1.98	0.11
E. virginicus	Exclusion	760.00	-3.70	-9.87	1.93	0.14
E. virginicus	Exclusion	1101.00	-1.69	-5.49	2.03	0.23
E. virginicus	Exclusion	1680.00	0.83	-1.56	3.36	0.71
E. virginicus	Exclusion	2433.00	3.20	-0.17	7.60	0.94
P. autumnalis	Access	873.00	-0.76	-5.08	3.35	0.35
P. autumnalis	Access	1122.00	-0.52	-3.80	2.75	0.38
P. autumnalis	Access	1443.00	-0.18	-2.60	2.28	0.45
P. autumnalis	Access	1923.00	0.31	-1.88	2.57	0.59
P. autumnalis	Access	2472.00	0.84	-2.35	4.34	0.67
P. autumnalis	Exclusion	873.00	2.73	-2.78	9.74	0.81
P. autumnalis	Exclusion	1122.00	2.96	-1.31	8.23	0.88
P. autumnalis	Exclusion	1443.00	3.13	0.02	6.92	0.95
P. autumnalis	Exclusion	1923.00	3.35	0.62	6.84	0.98
P. autumnalis	Exclusion	2472.00	3.46	-0.60	8.59	0.92

Model	Parameter	Median	Lower 95%	Upper 95%	P(>0)
Biomass (Gaussian)	Symbiont status (S+)	-0.001	-0.436	0.436	0.498
Biomass (Gaussian)	Symbiont status (S+) $\times$ Precipitation	0.010	-0.390	0.428	0.517
Inflo count (NB)	Symbiont status (S+)	-0.449	-1.007	0.104	0.054
Inflo count (NB)	Symbiont status (S+) $\times$ Precipitation	0.185	-0.388	0.719	0.757
Total spikelets projected (NB)	Symbiont status (S+)	-0.158	-0.550	0.233	0.208
Total spikelets projected (NB)	Symbiont status (S+) $\times$ Precipitation	-0.130	-0.459	0.184	0.214

Table S11: Greenhouse experiment: posterior summaries of the endophyte main effect (S^+) and its interaction with soil-origin precipitation for each fitness component. For each model, the table reports the median estimate, 95% credible interval (Lower and Upper), and the posterior probability that the effect is greater than zero (P($\beta > 0$)).

S.1.3 Supporting Figures

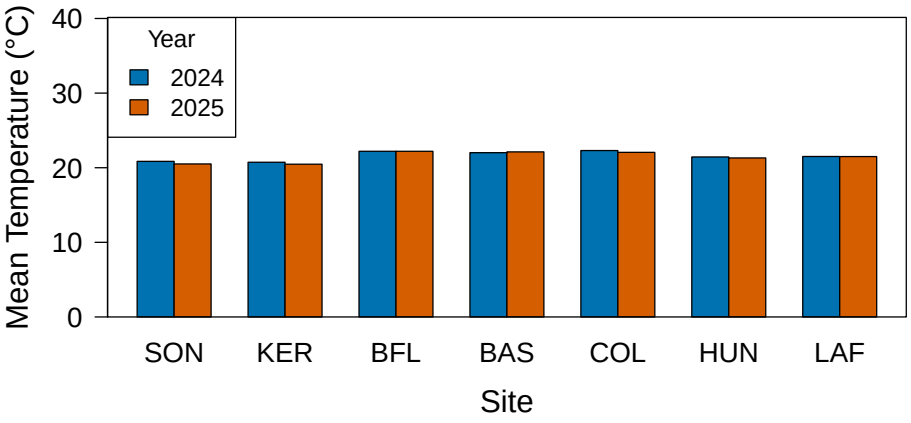


Fig S1: Annual temperature (°C) during the 2024 and 2025 census years. Sites are arranged west to east by longitude.

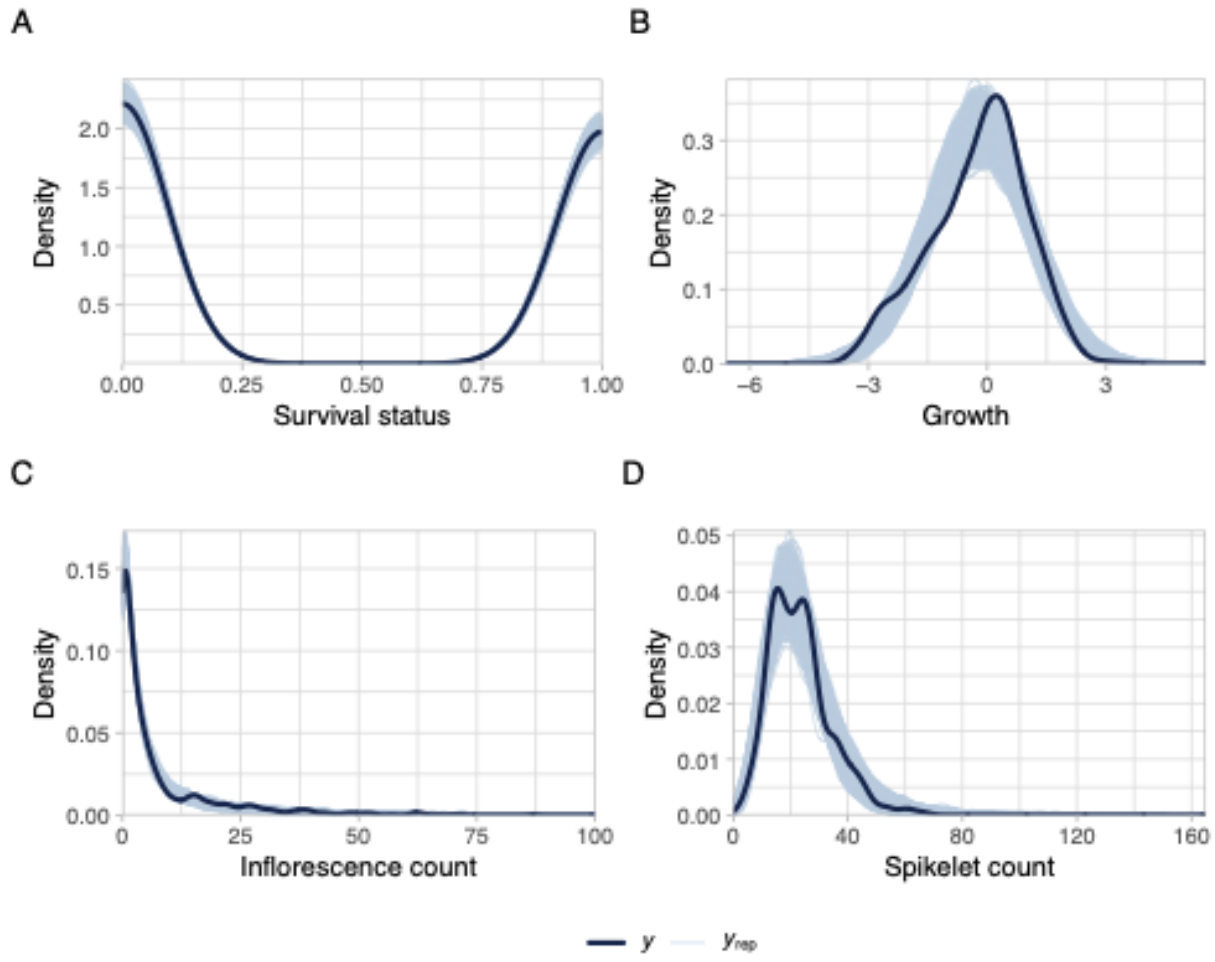


Fig S2: Posterior predictive checks for each vital rate. Dark blue lines show observed data (y) and light blue lines show replicated datasets from posterior predictions (y_{rep}). Strong overlap across survival, growth, inflorescence, and spikelet production indicates adequate model fit for all vital rates.

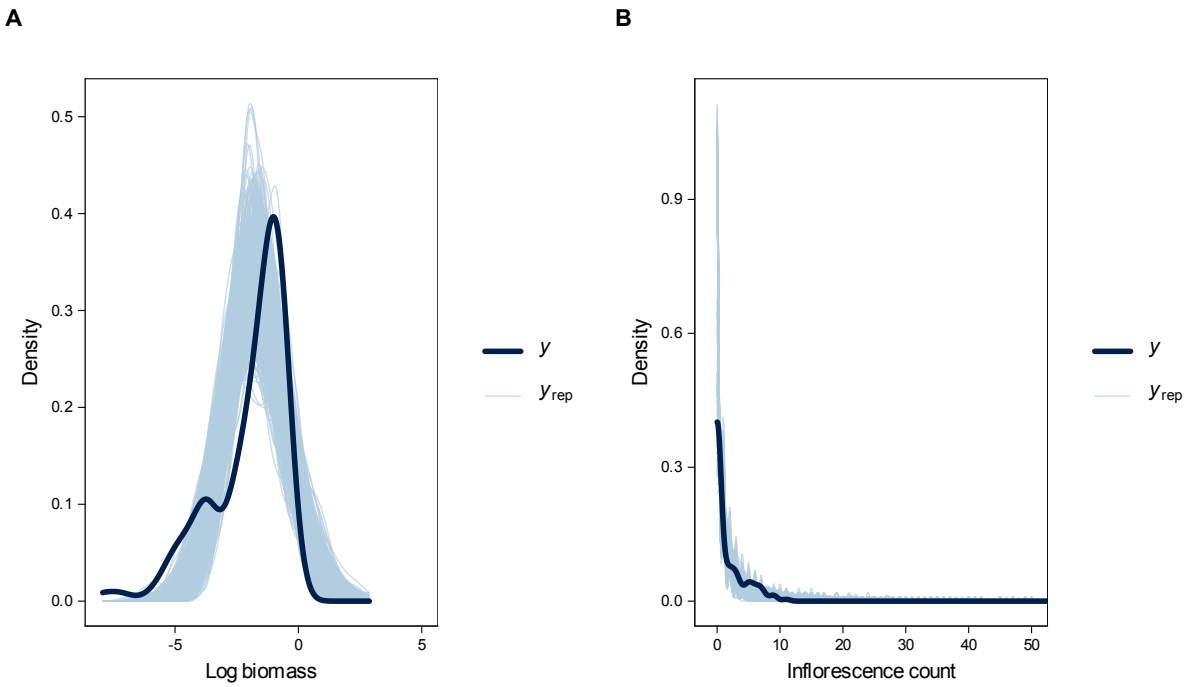


Fig S3: Posterior predictive checks for Bayesian models of plant performance in the greenhouse experiment. Density overlays show observed data (solid line) and 200 posterior predictive draws (shaded lines) for (A) aboveground biomass and (B) inflorescence count per plant. Models adequately capture the distribution of observed data across all response variables.

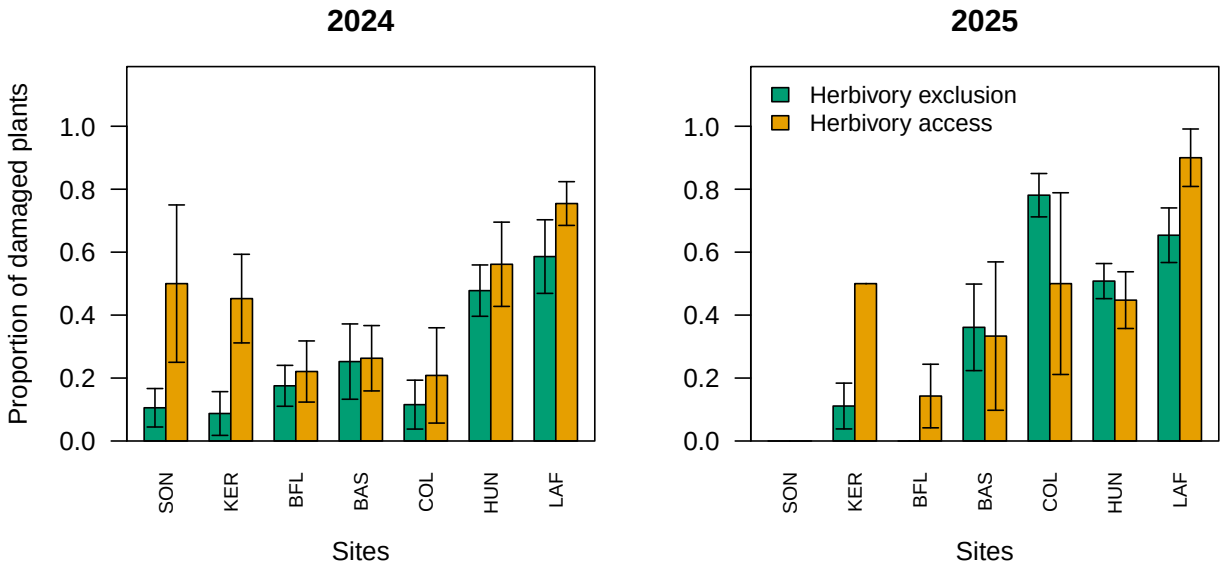


Fig S4: Proportion of plants damaged by herbivory across sites in 2024 and 2025. Sites are ordered west to east. Error bars indicate standard errors across plots within each site. Panels show results for 2024 (left) and 2025 (right).

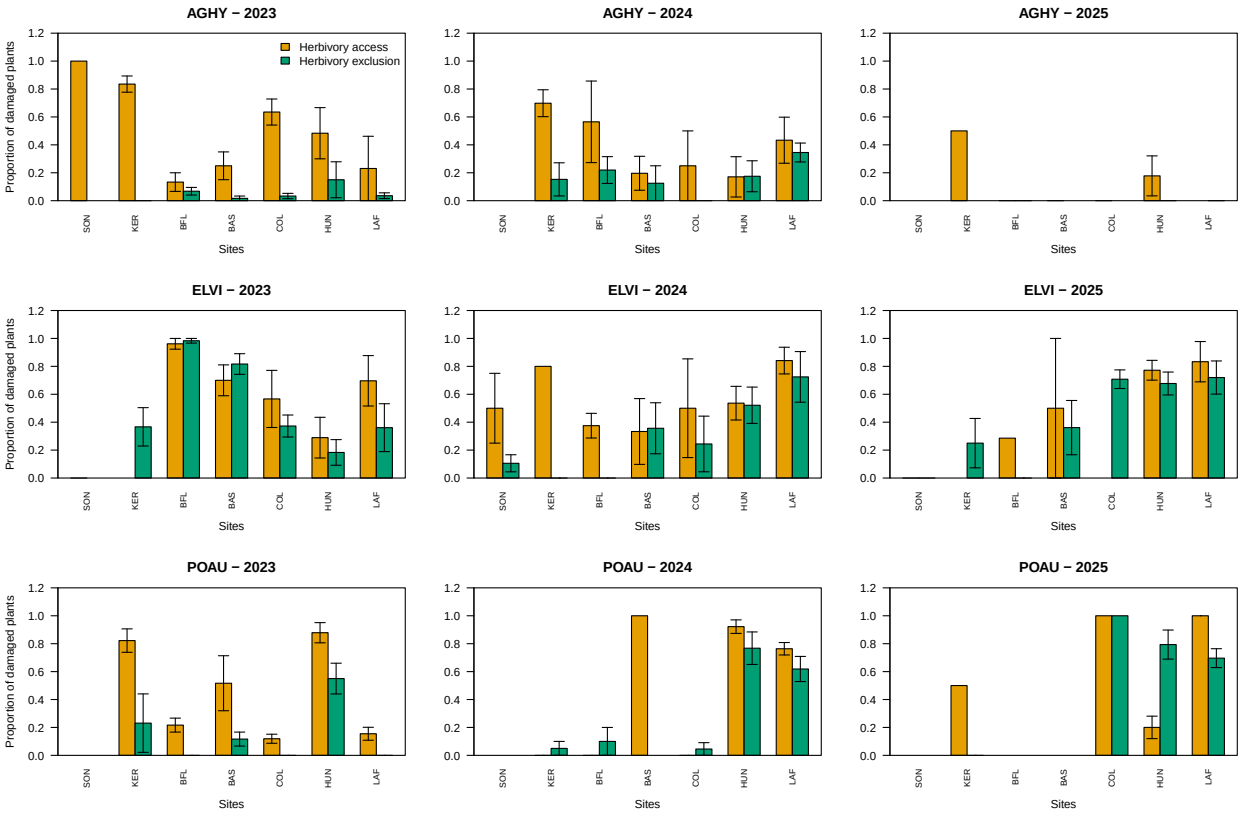


Fig S5: Proportion of plants damaged by herbivory across sites for *Agrostis hyemalis*, *Elymus virginicus*, and *Poa autumnalis* over 2023–2025. Sites are ordered west to east. Error bars indicate standard errors across plots within each site. Panels are arranged by species (rows) and year (columns).

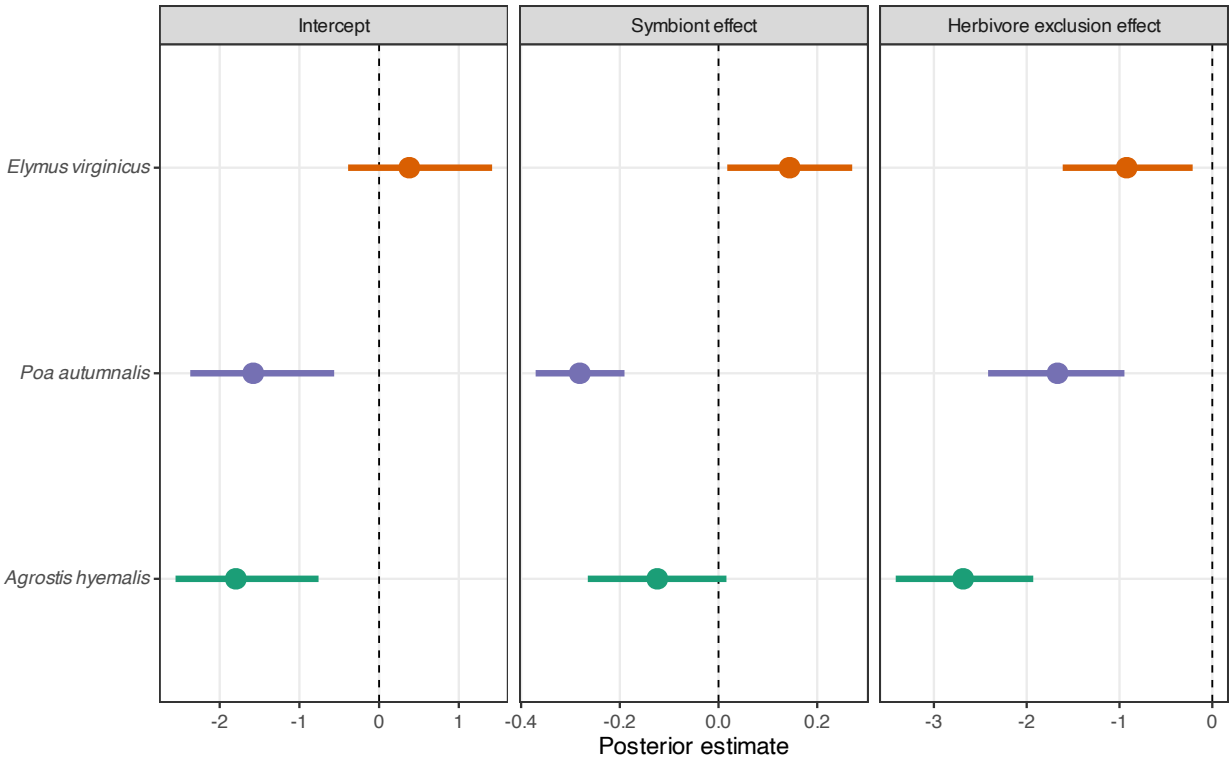


Fig S6: Posterior estimates (median and 90% credible intervals) of symbiont and herbivore exclusion effects on the proportion of damaged tillers for three host grass species, from a binomial model with site, year, and plot as random effects. Negative values indicate reduced herbivore damage, whereas positive values indicate increased damage. Back-transformed to predicted proportions of damaged tillers in herbivore-access plots, symbiosis reduced damage in *P. autumnalis* (S^+: 13.5% vs. S^-: 17.1%) and, more weakly, in *A. hyemalis* (S^+: 12.8% vs. S^-: 14.2%), and increased damage in *E. virginicus* (S^+: 62.8% vs. S^-: 59.4%). Points and intervals are colored by species: *Agrostis hyemalis* (green), *Elymus virginicus* (orange), and *Poa autumnalis* (purple). Dashed vertical lines indicate no effect.

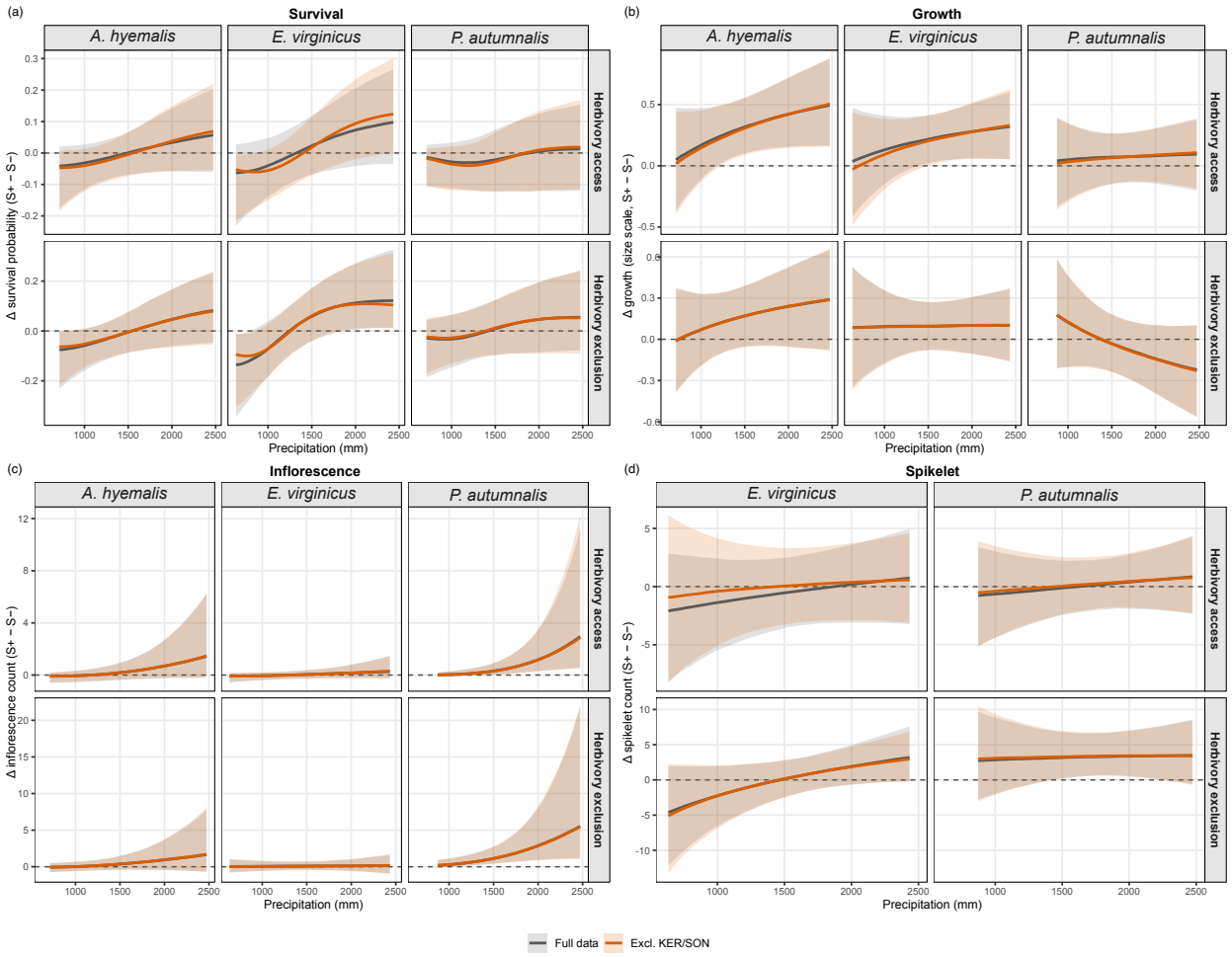


Fig S7: Sensitivity of estimated endophyte effects ($\Delta = S^{+} - S^{-}$) on (a) survival, (b) growth, (c) inflorescence production, and (d) spikelet production to exclusion of sites KER and SON, which were destroyed by deer and subsequently replanted in 2024. Lines and shaded bands show posterior median and 90% credible intervals for the full dataset (gray) and for the dataset excluding KER and SON (orange). Estimated effects were nearly identical between the two datasets across all species, vital rates, and herbivory treatments, indicating that the disturbance and replanting at these sites did not meaningfully influence our conclusions.

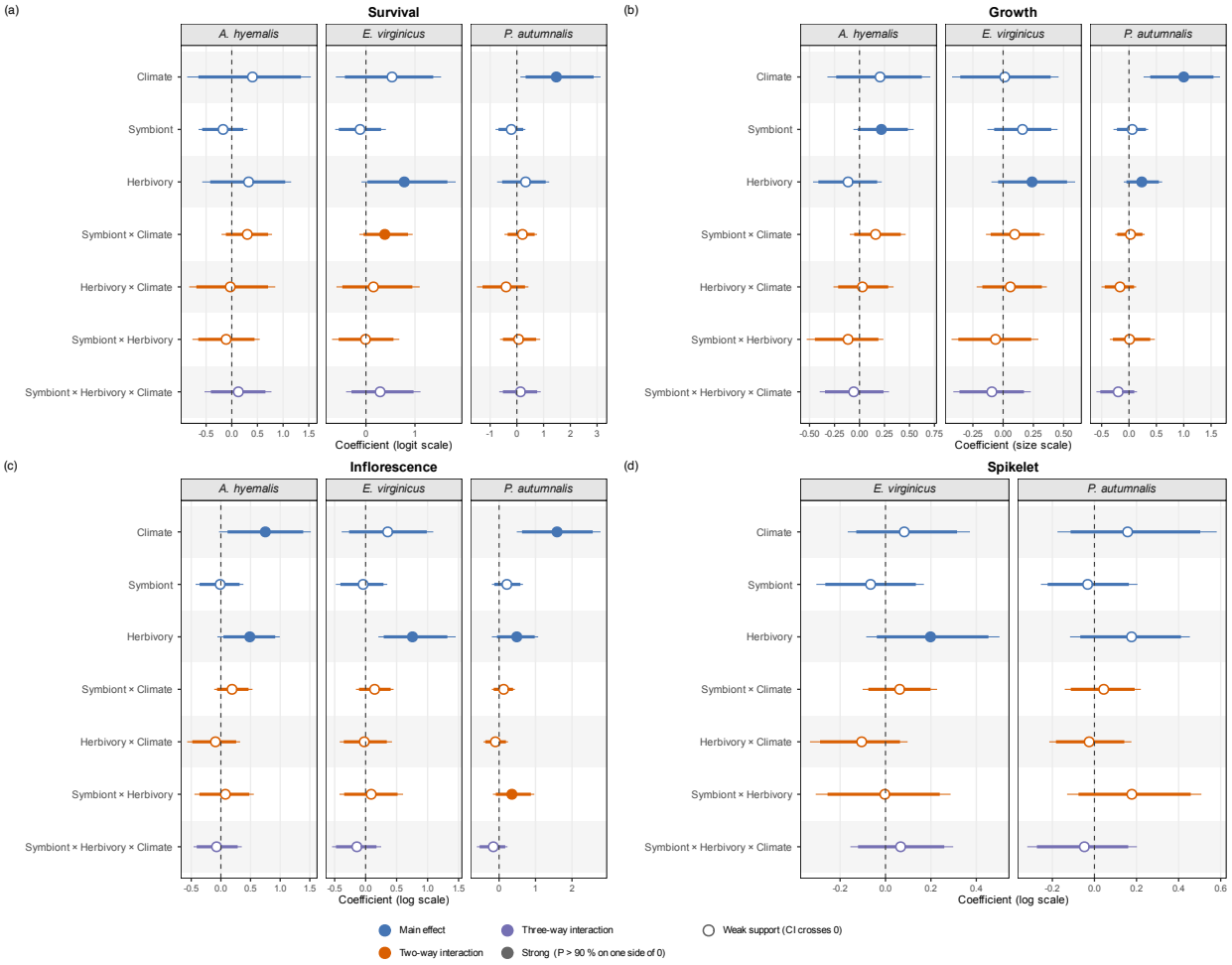


Fig S8: Posterior summaries of standardized regression coefficients from Bayesian models of survival, growth, inflorescence production, and spikelet production across three cool-season grass species (*Agrostis hyemalis*, *Elymus virginicus*, and *Poa autumnalis*; spikelet production was modeled for *E. virginicus* and *P. autumnalis* only). Panels show median posterior estimates (points), 90% credible intervals (thick bars), and 95% credible intervals (thin bars) for main effects of climate (precipitation), endophyte presence, and herbivory, as well as their two-way and three-way interactions. Filled points indicate strong posterior support for an effect ($\Pr(\beta > 0)$ or $\Pr(\beta < 0) > 0.90$), whereas open points indicate weaker support. Colors distinguish main effects, two-way interactions, and three-way interactions. Vertical dashed lines indicate no effect ($\beta = 0$). Coefficients are shown on the model link scale: logit for survival, linear size scale for growth, and log scale for inflorescence and spikelet production.

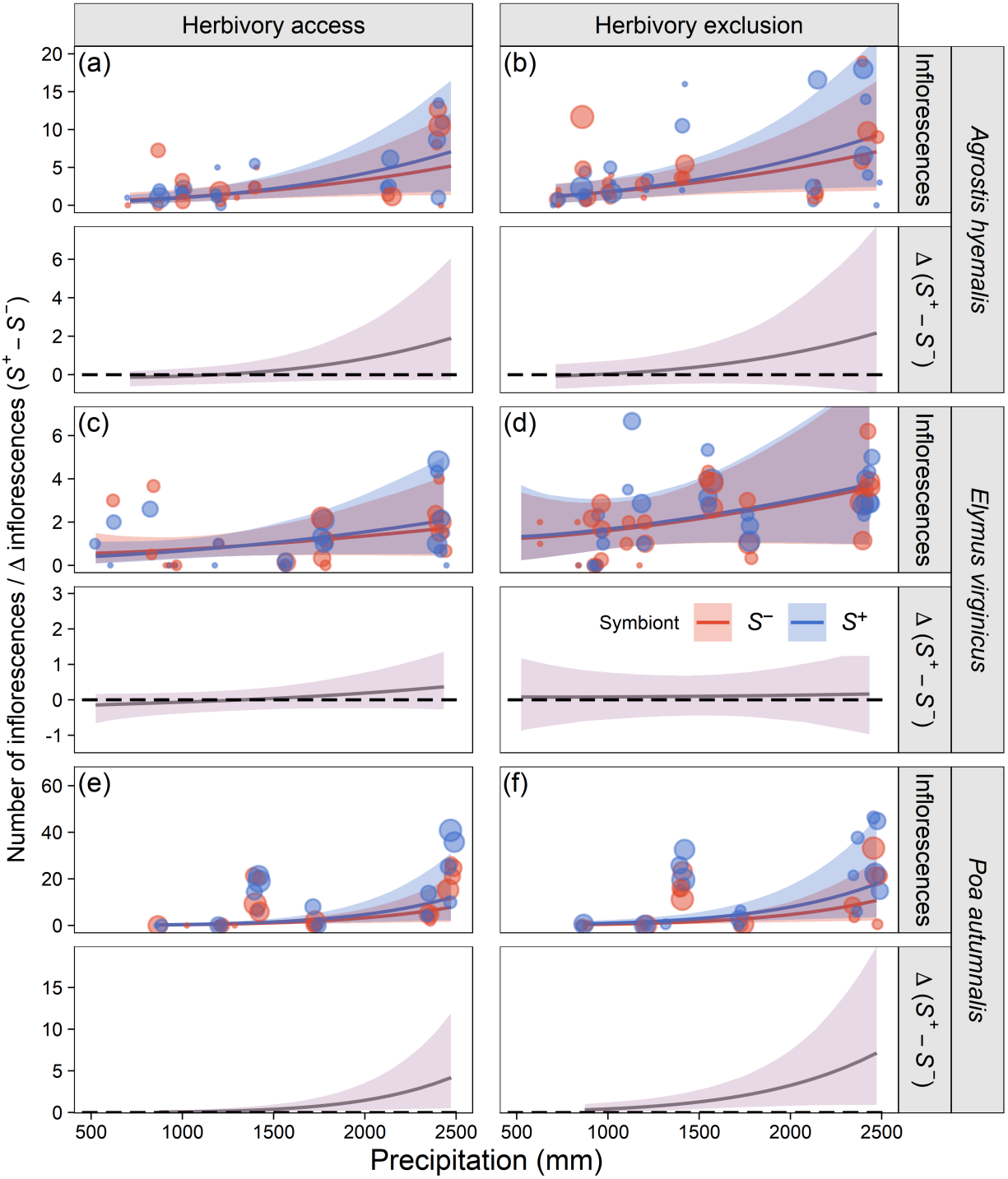


Fig S9: Predicted inflorescence production (upper panels) and symbiont effect ($\Delta = S^{+} - S^{-}$; lower panels) across the precipitation gradient. Lines show posterior means with 90% credible intervals. Points represent mean observed inflorescences per plot, slightly jittered for visualization. The dashed line at $\Delta = 0$ indicates no symbiont effect. Panels are arranged by species and herbivory treatment (herbivore access and exclusion). All panels share a common x-axis for illustrative purposes; some species extend beyond their observed precipitation range.

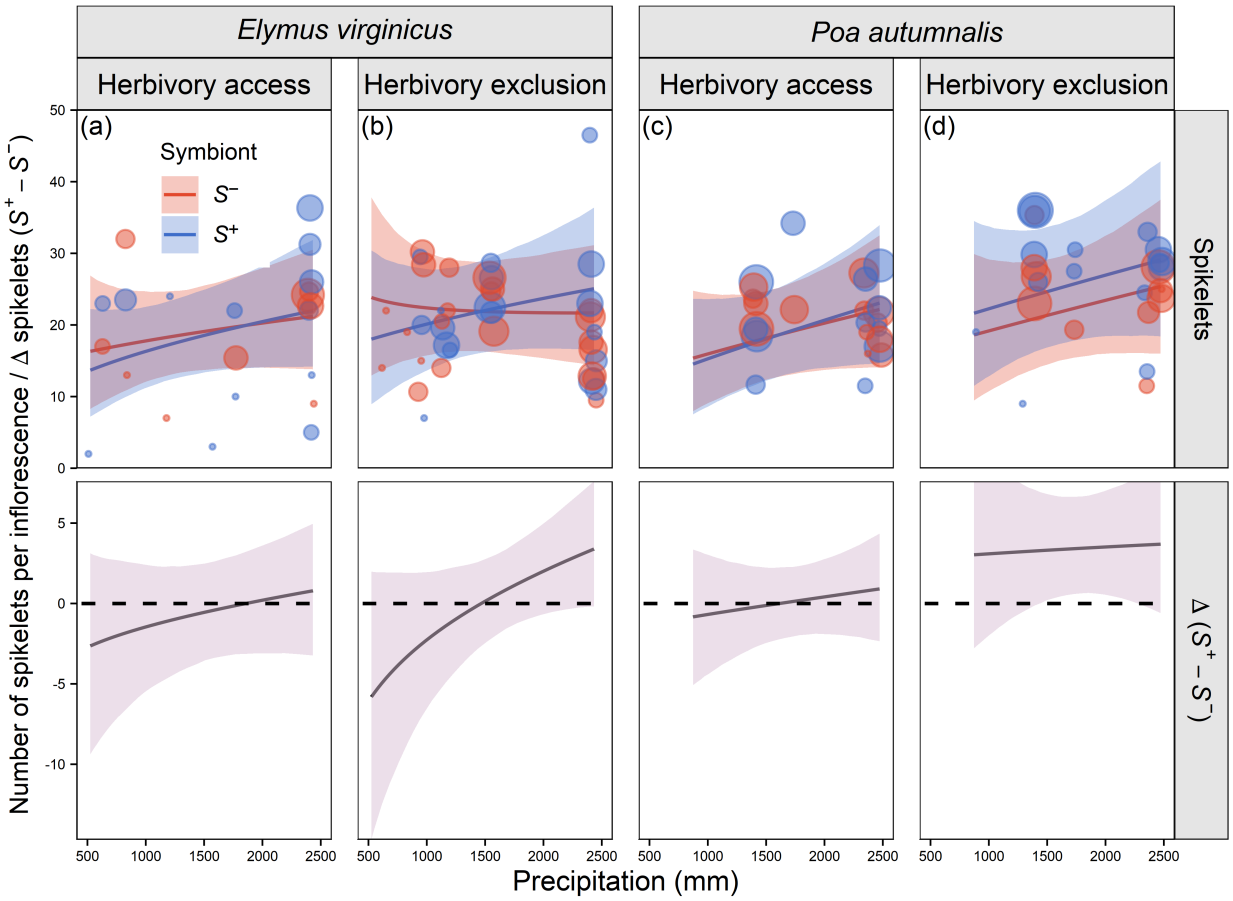


Fig S10: Predicted spikelet production (upper panels) and symbiont effect ($\Delta = S^{+} - S^{-}$; lower panels) across the precipitation gradient. Lines show posterior means with 90% credible intervals. Points represent mean observed spikelets per plot, slightly jittered for visualization. Spikelets were measured for two species only (see Methods). The dashed line at $\Delta = 0$ indicates no symbiont effect. Panels are arranged by species and herbivory treatment (herbivore access and exclusion). All panels share a common x-axis for illustrative purposes; some species extend beyond their observed precipitation range.

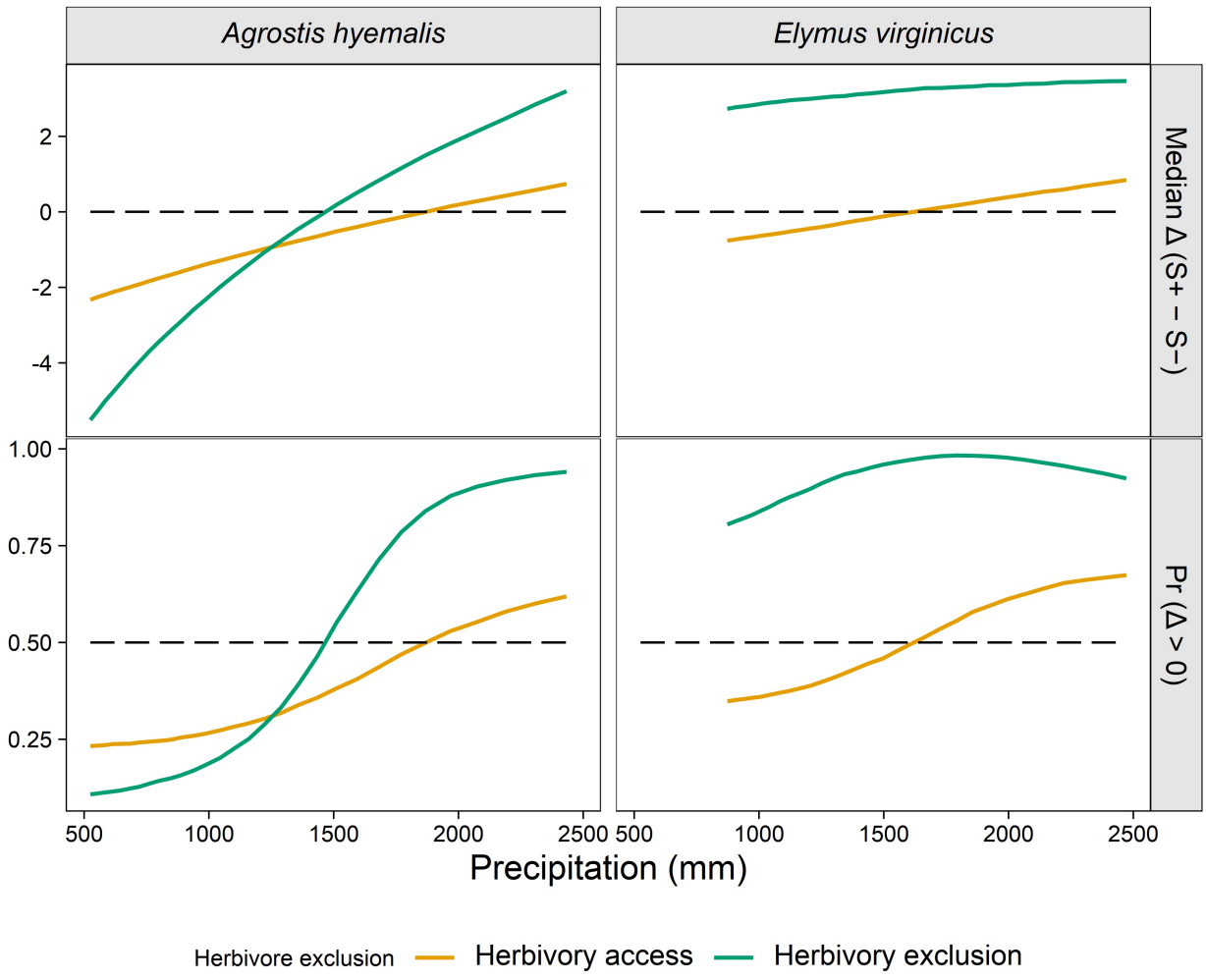


Fig S11: Median Δ ($S^+ - S^-$) and posterior probability $P(\Delta > 0)$ for spikelet production across precipitation quantiles for each species and herbivory treatment. Herbivore exclusion treatments are indicated by colour. Horizontal dashed lines indicate $\Delta = 0$ for medians and $P(\Delta > 0) = 0.5$ for probabilities. Panels are arranged by metric and species.

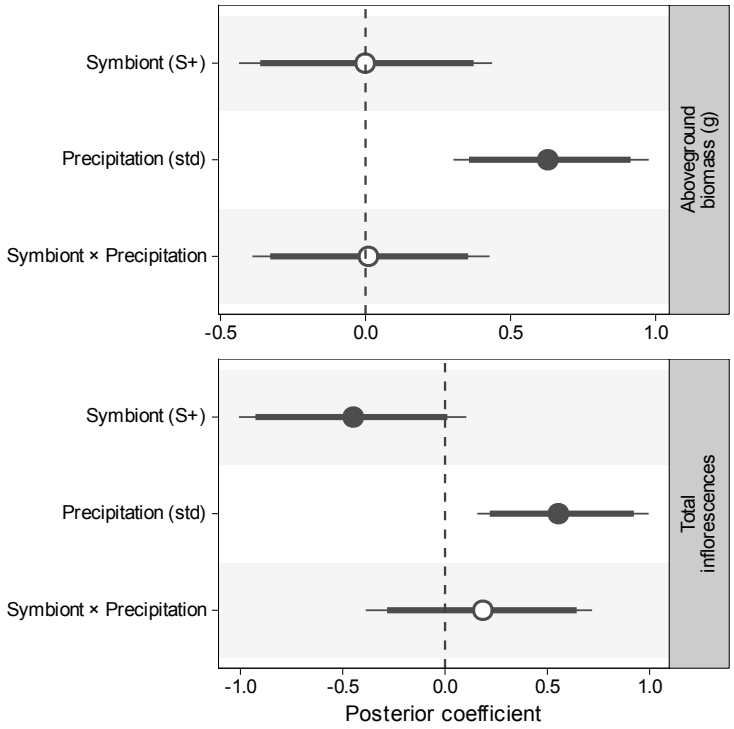


Fig S12: Posterior coefficient estimates for the greenhouse experiment (median, with thick and thin bars showing 90% and 95% credible intervals) for symbiont presence, soil-origin precipitation (standardized), and their interaction, for aboveground biomass (top) and total inflorescences (bottom). Filled points indicate estimates with posterior probability >0.9 of being above or below zero; the dashed line marks zero.

References

Fowler, J. C., Moutouama, J., and Miller, T. E. (2025). Increasing prevalence of plant-fungal symbiosis across two centuries of environmental change. *Global Change Biology*, 31(11):e70577.

Jackson, D. and Johnson-Cicalese, J. (1988). A rapid staining method for detection of endophytic fungi in turf and forage grasses. *Phytopathology*, 78:237–239.

Koh, S., Vicari, M., Ball, J., Rakocevic, T., Zaheer, S., Hik, D., and Bazely, D. (2006). Rapid detection of fungal endophytes in grasses for large-scale studies. *Functional Ecology*, pages 736–742.

Stan Development Team (2024). RStan: the R interface to Stan. R package version 2.32.6.